Cassiopea ornata

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Answer # 1 & 5 using data in your *species_name.len(genome)_summary.xlsx*

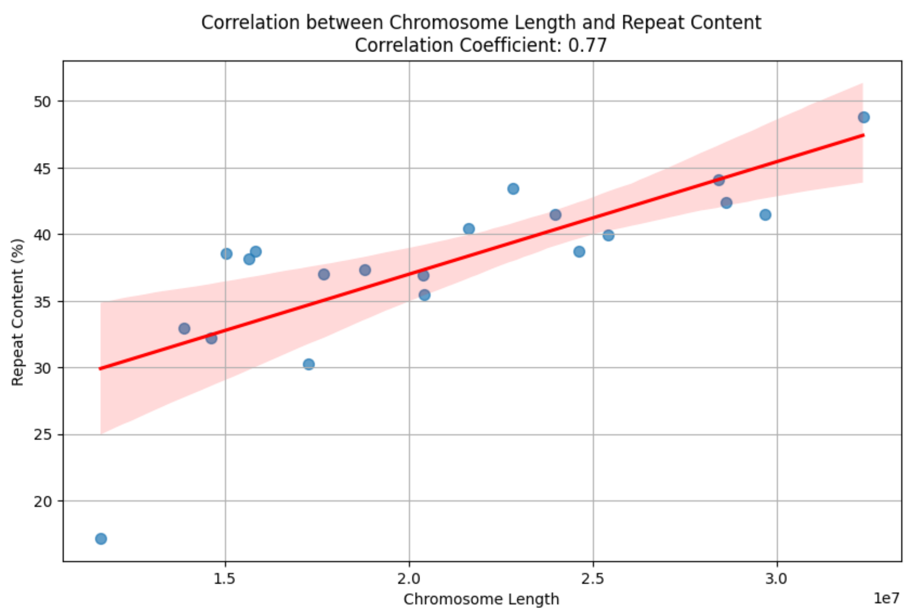
1. How many chromosomes in your species' genome assembly? (1 pts)
   a. 20 chromosomes
2. What is the total length of all the chromosomes combined? (1 pts)
   a. 48591587 bp
3. How many total scaffolds (chromosomes + unassigned scaffolds in the assembly?
   a. 490 scaffolds
4. What is the total length of the assembly (chromosomes + unassigned scaffolds)?
   a. 458652123 bp
5. What proportion of the assembly is included in the chromosomes (answer #2 / #4)? (1 pts)
   a. ~10.6%

Answer # 6 – 8 by comparing data in the highLevelCount.txt output files

6. Which chromosome has the highest repeat content? (2 pts)
   a. Chromosome ID: OZ199900.1, 48.8% repeat content
7. Which chromosome has the lowest repeat content? (2 pts)
   a. Chromosome ID: OZ199919.1 & Repeat content: 12.18%
8. Is there a correlation between the length of the chromosome and its repeat content? (8 pts)
   a. Report chromosome length and repeat content length for each chromosome

Chromosome_ID	Length	Repeat Content %
OZ199900.1	32329362	48.8
OZ199901.1	29664695	41.5
OZ199902.1	28619511	42.4
OZ199903.1	28396253	44.1
OZ199904.1	25413447	39.9
OZ199905.1	24600435	38.7
OZ199906.1	23970771	41.4
OZ199907.1	22809168	43.4

OZ199908.1	21628202	40.3
OZ199909.1	20401268	35.4
OZ199910.1	20391773	36.9
OZ199911.1	18787574	37.3
OZ199912.1	17668143	37.0
OZ199913.1	17280450	30.2
OZ199914.1	15815936	38.6
OZ199915.1	15654340	38.1
OZ199916.1	15022818	38.5
OZ199917.1	14631874	32.2
OZ199918.1	13879765	32.9
OZ199919.1	11625802	12.1

b. Report the correlation coefficient and p-value
   i. **Correlation coefficient r:** 0.77
   ii. **P-value:** 7.064*10^-5
   iii. 

c. Briefly describe your method for calculating the correlation
   i. We used Jupyter Notebook and Python to find the correlation coefficient by using pandas to create a data frame containing the chromosome ID, length, and repeat content, and the scipy.stats pearsonr package to find r and the p-value. We then used seaborn and matplotlib.pyplot to graph our results. From a manual approach, We calculated the mean for chromosome lengths (X) and repeat contents (Y), getting **20929579.35** for X and **37.485** for Y. We then computed the product of deviations and the squared deviations for X

and Y. Using these values, we calculated the Pearson correlation coefficient r=0.77, indicating a **strong positive correlation**.

Answer # 9 – 11 by comparing / combining information in the Earl Grey Summary files

9. Which TE has the highest copy number on each chromosome? (5 pts)
    a. Report chromosome ID, # of copies, *classification* (2 pts, Chr ID and copy number)
        i. +1 pt if only subclass name
        ii. +2 pts if subclass and superfamily
        iii. +3 pts if subclass, superfamily, and family

- OZ199900.1: 213 copies – Non-LTR retrotransposon, LINE, L1
- OZ199901.1: 312 copies – Non-LTR retrotransposon, PLE, Penelope
- OZ199902.1: 225 copies – Non-LTR retrotransposon, PLE, Penelope
- OZ199903.1: 235 copies – Non-LTR retrotransposon, LINE, L2
- OZ199904.1: 498 copies – Non-LTR retrotransposon, PLE, Penelope
- OZ199905.1: 200 copies – Non-LTR retrotransposon, PLE, Penelope
- OZ199906.1: 245 copies – Non-LTR retrotransposon, PLE, Penelope
- OZ199907.1: 345 copies – Non-LTR retrotransposon, LINE, L2
- OZ199908.1: 196 copies – Non-LTR retrotransposon, LINE, L2
- OZ199909.1: 172 copies – Non-LTR retrotransposon, PLE, Penelope
- OZ199910.1: 201 copies – Non-LTR retrotransposon, PLE, Penelope
- OZ199911.1: 262 copies – Non-LTR retrotransposon, PLE, Penelope
- OZ199912.1: 184 copies – Non-LTR retrotransposon, LINE, Penelope
- OZ199913.1: 139 copies – Non-LTR retrotransposon, PLE, Penelope
- OZ199914.1: 180 copies – Non-LTR retrotransposon, PLE, Penelope
- OZ199915.1: 268 copies – Non-LTR retrotransposon, PLE, Penelope,
- OZ199916.1: 248 copies – Non-LTR retrotransposon, PLE, Penelope
- OZ199917.1: 189 copies – DNA transposon, DNA, hAT
- OZ199918.1: 133 copies – Non-LTR retrotransposon, PLE, Penelope
- OZ199919.1: 139 copies – Non-LTR retrotransposon, LINE, L2

10. Which repeat *superfamily* comprises the most base pairs on each chromosome? (5 pts)
    a. Report chromosome ID, # of base pairs, *classification*, (2 pts, Chr ID and # of bases)

i. +1 pt if only subclass named
ii. +2 pts if subclass and superfamily
iii. +3 pts if subclass, superfamily, and family

OZ199900.1: 85686 - Non-LTR retrotransposon,LINE,R2-Hero

OZ199901.1: 149166 - Non-LTR retrotransposon,LINE,L2

OZ199902.1: 68229 - Non-LTR retrotransposon,LINE,Penelope

OZ199903.1: 131656 - Non-LTR retrotransposon,LINE,L2

OZ199904.1: 209245 - Non-LTR retrotransposon,LINE,Penelope

OZ199905.1: 100448 - Non-LTR retrotransposon,LINE,L2

OZ199906.1: 76780 - Non-LTR retrotransposon,LINE,CR1-Zenon

OZ199907.1: 156531 - Non-LTR retrotransposon,LINE,Penelope

OZ199908.1: 68883 - Non-LTR retrotransposon,LINE,L2

OZ199909.1: 167524 - DNA transposon,DNA,Maverick

OZ199910.1: 81452 - Non-LTR retrotransposon,LINE,Penelope

OZ199911.1: 91955 - Non-LTR retrotransposon,LINE,R2-Hero

OZ199912.1: 72385 - Non-LTR retrotransposon,LINE,L2

OZ199913.1: 47667 - Non-LTR retrotransposon,LINE,L2

OZ199914.1: 74651 - Non-LTR retrotransposon,LINE,CR1-Zenon

OZ199915.1: 93694 - Non-LTR retrotransposon,LINE,Penelope

OZ199916.1: 96549 - Non-LTR retrotransposon,LINE,Penelope

OZ199917.1: 75657 - Non-LTR retrotransposon,LINE,Penelope

OZ199918.1: 66872 - Non-LTR retrotransposon,LINE,R2-Hero

OZ199919.1: 90180 - Non-LTR retrotransposon,LINE,L2

Table specifying participation of each group member (6 pts)

Student Name	"Chromosome" Sequence ID assigned for analysis	Other contributions to project **(e.g. sequencing parsing, presenting development, canvas submission)**
1. Lauren Valenzuela	OZ199900.1 OZ199901.1 OZ199902.1 OZ199903.1	Sequence parsing and assigned chromosomes to group members, created excel spreadsheet deliverable, worked with Lorelei to answer questions 1-5, worked with group mates to answer 6-7 did math on my file for proportions and shared information. Uploaded TE landscapes to PowerPoint. Helped with understanding and walk through of how to answer question 9 and 10, I also adjusted some of the answers for the question as well. Added my ideas for how research would be continued for TE on the graduate student write up/reflection, my paragraph was the first paragraph. Shared excel file for submitting.
2. Lorelei Ing	OZ199904.1 OZ199905.1 OZ199906.1 OZ199907.1	Created Teams for communication, Answered questions 1-5 w Lauren, worked with group to answer 6 & 7 (everyone calculated the % proportion of their assigned chromosomes and we determined the highest and lowest from there), I then copied the excel we used for 6 & 7 into 8a, used R to

		create the TE content stacked bar plot after compiling every high level TE txt file into a csv, worked to troubleshoot the lack of landscape plots through singularity, created, shared, and helped write graduate student write up document—specifically talking about the future research ideas that this project could lead to, edited to get the word count down, submitted all project documents on behalf of the group to canvas
3. Mariah Noelle Cornelio	OZ199908.1 OZ199909.1 OZ199910.1	Created and shared PowerPoint slides and research summary word document, worked with groupmates to answer 6 and 7, created and filled in the table for 8A, worked with Darlene to find the correlation and p-value in 8B and 8C
4. Darlene Eligado	OZ199911.1 OZ199912.1 OZ199913.1	Worked with group makes to complete questions 6 and 7 to find the chromosome has the highest and lowest repeat content. Then collaborated with Mariah to complete 8B and 8C.
5. Nhi Tran	OZ199914.1 OZ199915.1 OZ199916.1	Answered question 9 for my group.
6. Pratyusha Gorige	OZ199917.1 OZ199918.1 OZ199919.1	Worked with my group to answer questions 6 and 7. Answered question 10 for my group. Added my thoughts to the graduate student write up doc.